



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005  
& ANSI/NCSL Z540-1-1994

ADVANCED TECHNOLOGY SERVICES  
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CALIBRATION

Valid to: March 31, 2017

Certificate Number: 1592.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations<sup>1</sup>:

I. Dimensional

| Parameter/Equipment                         | Range                      | CMC <sup>2,5</sup> (±)                                                                    | Comments                            |
|---------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------|-------------------------------------|
| Calipers                                    | Up to 6 in<br>(6 to 36) in | 22 $\mu\text{in}$ + 0.6R<br>64 $\mu\text{in}$ + 0.6R                                      | Gage blocks                         |
| Micrometers                                 | Up to 4 in<br>(4 to 36) in | 18 $\mu\text{in}$ + 0.6R<br>64 $\mu\text{in}$ + 0.6R                                      | Gage blocks                         |
| Height Gages                                | Up to 36 in                | 64 $\mu\text{in}$ + 0.6R                                                                  | Gage blocks                         |
| Dial Indicators                             | Up to 2 in                 | 34 $\mu\text{in}$ + 0.6R                                                                  | Optimar-100                         |
| Gage Blocks –<br>Ceramic<br>Chrome<br>Steel | Up to 4 in                 | (5.7 + 0.55L) $\mu\text{in}$<br>(4.8 + 0.88L) $\mu\text{in}$<br>(4.8 + 3L) $\mu\text{in}$ | Comparison to master<br>gage blocks |

*Peter M. Hays*

| Parameter/Equipment                                                     | Range                                     | CMC <sup>2, 5</sup> ( $\pm$ )               | Comments <sup>6</sup>                                                         |
|-------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------|
| Linear Displacement –<br>Incremental Gage<br>Linear Gages<br>Bore Gages | Up to 60 mm<br>Up to 60 mm<br>Up to 12 in | 0.22 $\mu$ m<br>0.21 $\mu$ m<br>28 $\mu$ in | Gage blocks<br>Pratt & Whitney <sup>TM</sup><br>Supermicrometer <sup>TM</sup> |
| Linear Standards –<br>Rings                                             | Up to 12 in                               | 21 $\mu$ in                                 | Pratt & Whitney<br>Labmaster <sup>TM</sup> UMS                                |
| Roundness                                                               | Up to 8.5 in                              | 91 $\mu$ in                                 | Federal Formtester                                                            |
| Attribute Gages –<br>Special Design Gages<br>(Fixture Gages), Form      | Up to 48 in<br>Up to 1200 mm              | 47 $\mu$ in<br>1.2 $\mu$ m                  | Leitz PMM                                                                     |
| Surface Roughness, Ra<br>Only                                           | Up to 500 $\mu$ m                         | 14 $\mu$ in                                 | Federal Surfanalyzer<br>5000 w/ roughness<br>speciman                         |
| Thread Plugs –<br>Pitch Diameter<br>up to 10 inches                     | (4 to 80) TPI<br>(0.35 to 30) TPmm        | 59 $\mu$ in<br>1.5 $\mu$ m                  | Pratt & Whitney<br>Supermicrometer <sup>TM</sup><br>Model C                   |
| Surface Plates                                                          | Up to (60 $\times$ 120) in                | (7.9 $\sqrt{DL}$ ) $\mu$ in                 | Electronic levels                                                             |
| Linear Standards <sup>5</sup> –<br>Pins, Plugs, Discs,<br>Lengths       | Up to 4 in<br>Up to 101 mm                | 21 $\mu$ in<br>0.53 $\mu$ m                 | Pratt & Whitney<br>Labmaster <sup>TM</sup> UMS                                |



## II. Electrical – DC/Low Frequency

| Parameter/Range       | Frequency                                                                                                                                          | CMC <sup>2,3</sup> (±)                                                                                                                                                              | Comments                  |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| AC Voltage – Generate |                                                                                                                                                    |                                                                                                                                                                                     |                           |
| (2.2 to 22) mV        | (1 to 20) Hz<br>(20 to 40) Hz<br>40 Hz to 20 kHz<br>(20 to 50) kHz<br>(50 to 100) kHz<br>(100 to 300) kHz<br>(300 to 500) kHz<br>500 kHz to 1 MHz  | 0.06 % + 6 $\mu$ V<br>0.02 % + 6 $\mu$ V<br>0.01 % + 6 $\mu$ V<br>0.04 % + 6.2 $\mu$ V<br>0.09 % + 8.4 $\mu$ V<br>0.13 % + 15 $\mu$ V<br>0.18 % + 30 $\mu$ V<br>0.48 % + 41 $\mu$ V | Fluke 5700A with<br>5725A |
| (22 to 220) mV        | (1 to 20) Hz<br>(20 to 40) Hz<br>40 Hz to 20 kHz<br>(20 to 50) kHz<br>(50 to 100) kHz<br>(100 to 300) kHz<br>(300 to 500) kHz<br>500 kHz to 1 MHz  | 0.06 % + 16 $\mu$ V<br>0.02 % + 10 $\mu$ V<br>0.01 % + 10 $\mu$ V<br>0.04 % + 10 $\mu$ V<br>0.09 % + 31 $\mu$ V<br>0.11 % + 32 $\mu$ V<br>0.18 % + 42 $\mu$ V<br>0.36 % + 0.11 mV   |                           |
| (0.220 to 2.2) V      | (1 to 20) Hz<br>20 Hz to 40 Hz<br>40 Hz to 20 kHz<br>(20 to 50) kHz<br>(50 to 100) kHz<br>(100 to 300) kHz<br>(300 to 500) kHz<br>500 kHz to 1 MHz | 0.06 % + 0.12 mV<br>0.02 % + 28 $\mu$ V<br>83 $\mu$ V/V + 14 $\mu$ V<br>0.14 mV/V + 30 $\mu$ V<br>0.03 % + 86 $\mu$ V<br>0.05 % + 0.17 mV<br>0.12 % + 0.41 mV<br>0.24 % + 1 mV      |                           |
| (2.2 to 22) V         | (1 to 20) Hz<br>20 Hz to 40 Hz<br>40 Hz to 20 kHz<br>(20 to 50) kHz<br>(50 to 100) kHz<br>(100 to 300) kHz<br>(300 to 500) kHz<br>500 kHz to 1 MHz | 0.06 % + 1 mV<br>0.18 mV/V + 0.34 mV<br>82 $\mu$ V/V + 0.15 mV<br>0.14 mV/V + 0.29 mV<br>0.03 % + 0.48 mV<br>0.06 % + 1.9 mV<br>0.14 % + 5.1 mV<br>0.30 % + 9.5 mV                  |                           |
| (22 to 220) V         | (1 to 20) Hz<br>20 Hz to 40 Hz<br>40 Hz to 20 kHz<br>(20 to 50) kHz<br>(50 to 100) kHz                                                             | 0.06 % + 10 mV<br>0.02 % + 3.3 mV<br>88 $\mu$ V/V + 1.7 mV<br>0.25 mV/V + 5 mV<br>0.06 % + 11 mV                                                                                    |                           |
| (220 to 1100) V       | 1 Hz to 1 kHz                                                                                                                                      | 86 $\mu$ V/V + 10 mV                                                                                                                                                                |                           |

*Peter Noyes*

| Parameter/Range                                 | Frequency                                                                         | CMC <sup>2, 3, 4</sup> ( $\pm$ )                                                                                     | Comments               |
|-------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------|
| AC Voltage – Measure,<br>Bandwidth $\leq$ 2 MHz |                                                                                   |                                                                                                                      |                        |
| 1 $\mu$ V to 12 mV                              | 1 Hz to 20 kHz<br>(20 to 50) kHz<br>(50 to 100) kHz<br>(100 to 300) kHz           | 0.013 % + 5.9 $\mu$ V<br>99 $\mu$ V/V + 71 $\mu$ V<br>0.19 % + 71 $\mu$ V<br>1.5 % + 560 $\mu$ V                     | HP 3458A               |
| 12 mV to 12 V                                   | 1 Hz to 20 kHz<br>(20 to 100) kHz<br>100 kHz to 1 MHz                             | 57 $\mu$ V/V + 2.2 mV<br>0.03 % + 11 mV<br>0.38 % + 140 mV                                                           |                        |
| (120 to 700) V                                  | 1 Hz to 20 kHz                                                                    | 0.026 % + 580 mV                                                                                                     |                        |
| AC Current – Generate                           |                                                                                   |                                                                                                                      |                        |
| Up To 220 $\mu$ A                               | (10 to 20) Hz<br>(20 to 40) Hz<br>40 Hz to 1 kHz<br>(1 to 5) kHz<br>(5 to 10) kHz | 0.10 % + 42 nA<br>0.06 % + 35 nA<br>0.03 % + 28 nA<br>0.10 % + 70 nA<br>0.25 % + 0.14 $\mu$ A                        | Fluke 5700A with 5725A |
| 220 $\mu$ A to 2.2 mA                           | (10 to 20) Hz<br>(20 to 40) Hz<br>40 Hz to 1 kHz<br>(1 to 5) kHz<br>(5 to 10) kHz | 0.11 % + 70 nA<br>0.06 % + 56 nA<br>0.02 % + 56 nA<br>0.10 % + 0.69 $\mu$ A<br>0.25 % + 1.4 $\mu$ A                  |                        |
| (2.2 to 22) mA                                  | (10 to 20) Hz<br>(20 to 40) Hz<br>40 Hz to 1 kHz<br>(1 to 5) kHz<br>(5 to 10) kHz | 0.11 % + 0.94 $\mu$ A<br>0.06 % + 0.62 $\mu$ A<br>0.02 % + 0.67 $\mu$ A<br>0.10 % + 7 $\mu$ A<br>0.25 % + 14 $\mu$ A |                        |
| (22 to 220) mA                                  | (10 to 20) Hz<br>(20 to 40) Hz<br>40 Hz to 1 kHz<br>(1 to 5) kHz<br>(5 to 10) kHz | 0.11 % + 9.4 $\mu$ A<br>0.06 % + 6.2 $\mu$ A<br>0.03 % + 6.5 $\mu$ A<br>0.10 % + 70 $\mu$ A<br>0.25 % + 0.14 mA      |                        |
| 220 mA to 2.2 A                                 | 20 Hz to 1 kHz<br>(1 to 5) kHz<br>(5 to 10) kHz                                   | 0.10 % + 65 $\mu$ A<br>0.12 % + 0.15 mA<br>1.4 % + 0.31 mA                                                           |                        |
| (2.2 to 11) A                                   | 40 Hz to 1 kHz<br>(1 to 5) kHz<br>(5 to 10) kHz                                   | 0.04 % + 1.2 mA<br>0.09 % + 0.97 mA<br>0.36 % + 0.99 mA                                                              |                        |

*Peter Nkomo*

| Parameter/Range      | Frequency                                           | CMC <sup>2,4</sup> (±)                                  | Comments |
|----------------------|-----------------------------------------------------|---------------------------------------------------------|----------|
| AC Current – Measure |                                                     |                                                         |          |
| 100 µA to 1 mA       | (10 to 45) Hz<br>45 Hz to 1 kHz                     | 0.035 % + 710 nA<br>0.014 % + 720 nA                    | HP 3458A |
| (1 to 100) mA        | (10 to 45) Hz<br>45 Hz to 10 kHz                    | 0.034 % + 72 µA<br>0.034 % + 72 µA                      |          |
| 100 mA to 1 A        | (10 to 45) Hz<br>(45 to 100) Hz<br>100 Hz to 10 kHz | 0.083 % + 1.6 mA<br>0.028 % + 1.7 mA<br>0.21 % + 1.5 mA |          |

| Parameter/Equipment   | Range                                                                                                    | CMC <sup>2,3,4</sup> (±)                                                                                              | Comments            |
|-----------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------|
| DC Voltage – Generate | Up to 220 mV<br>220 mV to 2.2 V<br>(2.2 to 11) V<br>(11 to 22) V<br>(22 to 220) V<br>(220 to 1100) V     | 15 µV/V + 0.93 µV<br>14 µV/V + 0.26 µV<br>6.2 µV/V + 29 µV<br>11 µV/V + 23 µV<br>15 µV/V + 84 µV<br>9.8 µV/V + 2.4 mV | Fluke 5700A         |
| DC Voltage – Measure  | Up to 100 mV<br>100 mV to 1 V<br>(1 to 10) V<br>(10 to 100) V<br>(100 to 1000) V                         | 1.7 µV/V<br>0.08 % + 30 µV<br>0.06 µV/V + 58 µV<br>0.15 µV/V + 390 µV<br>0.96 µV/V + 55 µV                            | HP 3458A            |
| DC Current – Generate | Up to 220 µA<br>(0.22 to 2.2) mA<br>(2.2 to 22) mA<br>(22 to 220) mA<br>(0.22 to 2.2) A<br>(2.2 to 11) A | 56 µA/A + 12 nA<br>60 µA/A + 52 nA<br>58 µA/A + 0.2 µA<br>68 µA/A + 1.7 µA<br>92 µA/A + 40 µA<br>0.03 % + 1 mA        | Fluke 5700A + 5725A |
| DC Current – Measure  | (10 to 100) µA<br>100 µA to 1 mA<br>(1 to 10) mA<br>(10 to 100) mA<br>100 mA to 1 A                      | 1.4 µA/A + 2.3 nA<br>12 µA/A + 19 nA<br>12 µA/A + 19 nA<br>17 µA/A + 3.9 µA<br>44 µA/A + 150 µA                       | HP 3458A            |

*Peter Nkomo*

| Parameter/Equipment                       | Range                                                                                                                                                                                                                                                 | CMC <sup>2, 3, 4</sup> (±)                                                                                                                                                                                             | Comments    |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Capacitance – Generate                    | (0.19 to 0.39999) nF<br>(0.4 to 1.0999) nF<br>(1.1 to 3.2999) nF<br>(3.3 to 10.9999) nF<br>(11 to 32.9999) nF<br>(33 to 109.999) nF<br>(110 to 329.999) nF<br>(0.33 to 1.09999) μF<br>(1.1 to 3.2999) μF<br>(3.3 to 10.9999) μF<br>(11 to 32.9999) μF | 0.66 % + 0.041 nF<br>0.36 % + 0.016 nF<br>0.46 % + 0.015 nF<br>0.56 % + 0.01 nF<br>0.25 % + 0.11 nF<br>0.62 % + 0.02 nF<br>0.27 % + 0.41 nF<br>0.22 % + 1.6 nF<br>0.27 % + 3.1 nF<br>0.29 % + 9.5 nF<br>0.42 % + 30 nF | Fluke 5520A |
| DC Resistance –<br>Generate, Fixed Points | 0 Ω<br>1 Ω<br>1.9 Ω<br>10 Ω<br>19 Ω<br>100 Ω<br>190 Ω<br>1 kΩ<br>1.9 kΩ<br>10 kΩ<br>19 kΩ<br>100 kΩ<br>190 kΩ<br>1 MΩ<br>1.9 MΩ<br>10 MΩ<br>19 MΩ<br>100 MΩ                                                                                           | 0.15 mΩ<br>0.18 mΩ<br>0.31 mΩ<br>0.40 mΩ<br>0.74 mΩ<br>2.2 mΩ<br>4.2 mΩ<br>16 mΩ<br>32 mΩ<br>170 mΩ<br>290 mΩ<br>1.7 Ω<br>3.3 Ω<br>25 Ω<br>51 Ω<br>500 Ω<br>1.3 kΩ<br>17 kΩ                                            | Fluke 5700A |
| DC Resistance –<br>Measure                | (0 to 10) Ω<br>(10 to 100) Ω<br>(0.1 to 1.0) kΩ<br>(1.0 to 10) kΩ<br>(10 to 100) kΩ<br>(0.1 to 1.0) MΩ<br>(1.0 to 10) MΩ<br>(10 to 100) MΩ<br>100 MΩ to 1.0 GΩ                                                                                        | 21 μΩ/Ω + 76 μΩ<br>17 μΩ/Ω + 0.76 mΩ<br>14 μΩ/Ω + 0.75 mΩ<br>14 μΩ/Ω + 7.4 mΩ<br>14 μΩ/Ω + 75 mΩ<br>21 μΩ/Ω + 3.8 Ω<br>74 μΩ/Ω + 150 Ω<br>0.072 % + 1.3 kΩ<br>0.70 % + 2.3 kΩ                                          | HP 3458A    |

*Peter Noyes*

| Parameter/Equipment                     | Range                                                                                             | CMC <sup>2</sup> (±)                                | Comments    |
|-----------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------|-------------|
| Electrical Simulation of Thermocouple – |                                                                                                   |                                                     |             |
| Type E                                  | (-250 to -100) °C<br>(-100 to -25) °C<br>(-25 to 350) °C<br>(350 to 650) °C<br>(650 to 1000) °C   | 0.51 °C<br>0.16 °C<br>0.14 °C<br>0.16 °C<br>0.21 °C | Fluke 5520A |
| Type J                                  | (-210 to -100) °C<br>(-100 to -30) °C<br>(-30 to 150) °C<br>(150 to 760) °C<br>(760 to 1200) °C   | 0.27 °C<br>0.16 °C<br>0.14 °C<br>0.17 °C<br>0.23 °C |             |
| Type K                                  | (-200 to -100) °C<br>(-100 to -25) °C<br>(-25 to 120) °C<br>(120 to 1000) °C<br>(1000 to 1372) °C | 0.34 °C<br>0.19 °C<br>0.17 °C<br>0.26 °C<br>0.40 °C |             |
| Type N                                  | (-200 to -100) °C<br>(-100 to -25) °C<br>(-25 to 120) °C<br>(120 to 410) °C<br>(410 to 1300) °C   | 0.40 °C<br>0.22 °C<br>0.19 °C<br>0.18 °C<br>0.27 °C |             |
| Type R                                  | (0 to 250) °C<br>(250 to 1000) °C<br>(1000 to 1400) °C<br>(1400 to 1767) °C                       | 0.66 °C<br>0.36 °C<br>0.37 °C<br>0.45 °C            |             |
| Type S                                  | (0 to 250) °C<br>(250 to 1000) °C<br>(1000 to 1400) °C<br>(1400 to 1767) °C                       | 0.61 °C<br>0.40 °C<br>0.39 °C<br>0.47 °C            |             |
| Type T                                  | (-250 to -150) °C<br>(-150 to 0) °C<br>(0 to 120) °C<br>(120 to 400) °C                           | 0.64 °C<br>0.24 °C<br>0.16 °C<br>0.14 °C            |             |



| Parameter/Equipment       | Range                                                                                                                                                                  | CMC <sup>2</sup> (±)                                                                           | Comments    |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------|
| RTD Simulation – Generate |                                                                                                                                                                        |                                                                                                |             |
| Pt 385, 100 Ω             | (-200 to 0) °C<br>(0 to 100) °C<br>(100 to 300) °C<br>(300 to 400) °C<br>(400 to 630) °C<br>(630 to 800) °C                                                            | 0.05 °C<br>0.07 °C<br>0.09 °C<br>0.1 °C<br>0.12 °C<br>0.23 °C                                  | Fluke 5520A |
| Pt 385, 200 Ω             | (-200 to 100) °C<br>(100 to 260) °C<br>(260 to 300) °C<br>(300 to 400) °C<br>(400 to 600) °C<br>(600 to 630) °C                                                        | 0.04 °C<br>0.051 °C<br>0.12 °C<br>0.13 °C<br>0.14 °C<br>0.16 °C                                |             |
| Pt 385, 500 Ω             | (-200 to -80) °C<br>(-80 to 100) °C<br>(100 to 260) °C<br>(260 to 300) °C<br>(300 to 400) °C<br>(400 to 600) °C<br>(600 to 630) °C                                     | 0.04 °C<br>0.05 °C<br>0.06 °C<br>0.08 °C<br>0.081 °C<br>0.091 °C<br>0.12 °C                    |             |
| Pt 385, 1000 Ω            | (-200 to 0) °C<br>(0 to 100) °C<br>(100 to 260) °C<br>(260 to 300) °C<br>(300 to 600) °C<br>(600 to 630) °C                                                            | 0.03 °C<br>0.04 °C<br>0.05 °C<br>0.061 °C<br>0.071 °C<br>0.23 °C                               |             |
| Pt 3916, 100 Ω            | (-200 to -190) °C<br>(-190 to -80) °C<br>(-80 to 0) °C<br>(0 to 100) °C<br>(100 to 260) °C<br>(260 to 300) °C<br>(300 to 400) °C<br>(400 to 600) °C<br>(600 to 630) °C | 0.25 °C<br>0.04 °C<br>0.05 °C<br>0.06 °C<br>0.07 °C<br>0.08 °C<br>0.09 °C<br>0.1 °C<br>0.23 °C |             |



| Parameter/Equipment                 | Range                                                                                                       | CMC <sup>2</sup> (±)                                          | Comments    |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------|
| RTD Simulation –<br>Generate (cont) |                                                                                                             |                                                               |             |
| Pt 3926, 100 Ω                      | (-200 to -80) °C<br>(-80 to 0) °C<br>(0 to 100) °C<br>(100 to 300) °C<br>(300 to 400) °C<br>(400 to 630) °C | 0.05 °C<br>0.05 °C<br>0.07 °C<br>0.09 °C<br>0.1 °C<br>0.12 °C | Fluke 5520A |
| Pt Ni 385, 120 Ω<br>(NI120)         | (-80 to 100) °C<br>(100 to 260) °C                                                                          | 0.08 °C<br>0.14 °C                                            |             |
| Cu 427, 10 Ω                        | (-100 to 260) °C                                                                                            | 0.3 °C                                                        |             |

### III. Mechanical

| Parameter/Equipment             | Range                                                                                                       | CMC <sup>2</sup> (±)                                                                                                         | Comments                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Torque – Measuring<br>Equipment | (0.28 to 2.8) N·m<br>(0.56 to 12.4) N·m<br>(1.1 to 339) N·m<br>(4.5 to 1 355) N·m                           | 0.011 % rdg + 0.0011 N·m<br>0.048 % rdg + 0.00076 N·m<br>0.0019 % rdg + 0.14 N·m<br>0.034 % rdg + 0.13 N·m                   | Arms and weights             |
| Torque – Measure                | (1.1 to 5.9) N·m<br>(5.4 to 28.0) N·m<br>(22.5 to 112.7) N·m<br>(67.2 to 338.2) N·m<br>(269.4 to 1 353) N·m | 0.22 % rdg + 0.0020 N·m<br>0.33 % rdg + 0.048 N·m<br>0.30 % rdg + 0.14 N·m<br>0.21 % rdg + 0.25 N·m<br>0.21 % rdg + 0.96 N·m | Torque CDI measure<br>system |
| Force – Measure                 | (0 to 150) lbf                                                                                              | 0.10 lbf                                                                                                                     | Adapters and<br>weights      |

### IV. Time & Frequency

| Parameter/Equipment                | Range                                                                                                                     | CMC <sup>2</sup> (±)                                                                                                    | Comments    |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------|
| Frequency – Measuring<br>Equipment | (1.0 to 119.990) Hz<br>(120.00 to 1 199.90) Hz<br>(1.2 to 11.9990) kHz<br>(12.0 to 119.990) kHz<br>(120.0 to 1199.90) kHz | 6.3 mHz + 4.1 μHz/Hz<br>58 mHz + 0.27 μHz/Hz<br>4.0 mHz + 1.2 μHz/Hz<br>9.3 mHz + 0.083 μHz/Hz<br>58 Hz + 0.0027 μHz/Hz | Fluke 5520A |

<sup>1</sup>This laboratory offers commercial calibration service.

<sup>2</sup> Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of  $k = 2$ . The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

<sup>3</sup> The measurands stated are generated with the Fluke 5700A series of instruments. This capability is suitable for the calibration of the devices intended to measure the stated measurand in the ranges indicated. Calibration and Measurement Capability is expressed as either a specific value that covers the full range or as a fraction percent of the reading plus a fixed floor specification.

<sup>4</sup> The measurands stated are measured with the HP 3458A. This capability is suitable for the calibration of the devices intended to generate the measurand in the ranges indicated. Calibration and Measurement Capability is expressed as either a specific value that covers the full range or as a combination of the fraction percent of the reading/output plus a range specification.

<sup>5</sup> In the statement of CMC,  $L$  is the numerical value of the nominal length of the device measured in inches. In the statement of CMC,  $R$  is the resolution of the unit under test in inches. In the statement of CMC for Surface Plates,  $DL$  is the numerical value of the diagonal.

<sup>6</sup> "Labmaster" and "Supermicrometer" are registered trademarks owned by Pratt & Whitney Measurement Systems, Inc., Connecticut U.S.A.





## ***Accredited Laboratory***

A2LA has accredited

### **ADVANCED TECHNOLOGY SERVICES**

*Peoria, IL*

for technical competence in the field of

### **Calibration**

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANSI/NCSL Z540-1-1994 and any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009).



Presented this 7<sup>th</sup> day of July 2015.

A handwritten signature in black ink, reading 'Peter Meyer'.

President & CEO  
For the Accreditation Council  
Certificate Number 1592.01  
Valid to March 31, 2017

*For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.*